

CENG381 Stochastic Processes

Renewal-Reward and Residual Life — Practice 1 (Answer Key)

Q1 Solution

a) Deterministic $X=6$:

$$E[X] = 6, \quad E[X^2] = 6^2 = 36$$
$$Y = 36 / (2 \cdot 6) = 3 = E[X]/2 \quad \checkmark$$

Geometrically: between two renewals at 0 and 6, the residual life $Y(t) = 6-t$ decreases linearly from 6 to 0. The time average of a decreasing line from 6 to 0 is exactly $3 = 6/2$.

b) Exponential $X \sim \text{Exp}(1/6)$:

$$E[X] = 6, \quad E[X^2] = 2/\lambda^2 = 2/(1/6)^2 = 2 \cdot 36 = 72$$
$$Y = 72 / (2 \cdot 6) = 6 = E[X]$$

$Y = E[X]$ rather than $E[X]/2$ because the exponential is memoryless. At any time t , the remaining life has the same $\text{Exp}(1/6)$ distribution as the full lifetime X — the distribution does not depend on how long you have already waited. This is the defining property of the exponential.

c) Uniform $[0,12]$:

$$E[X] = 6, \quad E[X^2] = (\theta^2 + \theta \cdot 12 + 12^2)/3 = 144/3 = 48$$
$$Y = 48 / (2 \cdot 6) = 4 > E[X]/2 = 3$$

$Y > E[X]/2$: this is the inspection paradox. If you arrive at a random time, you are more likely to land inside a long inter-renewal interval than a short one (long intervals cover more time). On average the interval you land in is longer than a typical interval, and therefore the expected remaining wait exceeds half the mean interval length.

Q2 Solution

a) Two-point distribution ($X = 1$ w.p. $1/2$, $X = 5$ w.p. $1/2$):

$$E[X] = (1+5)/2 = 3$$
$$E[X^2] = (1^2 + 5^2)/2 = (1+25)/2 = 13$$
$$E[L] = E[X^2]/E[X] = 13/3 \approx 4.33$$

$$\text{Excess: } E[L] - E[X] = 13/3 - 3 = 4/3 \approx 1.33 \text{ (absolute)}$$
$$E[L]/E[X] = (13/3)/3 = 13/9 \approx 1.44 \text{ (44\% longer)}$$

The interval of length 5 is 5 times longer than the interval of length 1, so it 'catches' 5 times as many

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random arrivals. The observed interval is heavily biased towards the longer type.

b) Exponential $X \sim \text{Exp}(1/6)$:

$$\begin{aligned}E[X] &= 6, & E[X^2] &= 72 \\E[L] &= 72/6 = 12 = 2 \cdot E[X]\end{aligned}$$

The observed interval is twice as long as the average interval. Interpretation: you arrive in an interval of length X , with your position uniform in $[0, X]$ (since arrivals are Poisson/memoryless). Given you are in interval X , your remaining wait is $\text{Exp}(1/6)$ with mean 6, and the time already elapsed is also $\text{Exp}(1/6)$ with mean 6 — making the total interval length $12 = 2 \cdot E[X]$ on average.

c) Erlang(2, 1/3):

$$\begin{aligned}E[X] &= 6, & \text{Var}(X) &= 18, & E[X^2] &= 18 + 36 = 54 \\E[L] &= 54/6 = 9\end{aligned}$$

Erlang $E[L]=9$ vs Exp $E[L]=12$ (both mean 6). The Erlang has smaller variance (18 vs 36 for Exp), so the inspection paradox is less extreme. In general: $E[L] = E[X] + \text{Var}(X)/E[X]$. Less variability $\rightarrow$ smaller $\text{Var}(X)/E[X] \rightarrow$ observed interval closer to the true mean.

Q3 Solution

a) $E[X^2]$ and time-average residual life:

$$\begin{aligned}X &\sim \text{Uniform}[4, 16]: \quad a=4, \quad b=16 \\E[X] &= (4+16)/2 = 10 \\E[X^2] &= (a^2 + a \cdot b + b^2)/3 = (16 + 64 + 256)/3 = 336/3 = 112 \\Y &= E[X^2] / (2 \cdot E[X]) = 112 / 20 = 5.6 \text{ minutes}\end{aligned}$$

b) Comparison with deterministic and connection to variance:

$$\begin{aligned}\text{Deterministic } (X=10): \quad Y_{\text{det}} &= 10/2 = 5.0 \text{ minutes} \\ \text{Random } (U[4,16]): \quad Y_{\text{rnd}} &= 5.6 \text{ minutes} \\ \text{Excess} &= 5.6 - 5.0 = 0.6 \text{ minutes} \\ \\ \text{Var}(X) &= (b-a)^2/12 = 144/12 = 12 \\ \text{Excess} &= \text{Var}(X) / (2 \cdot E[X]) = 12/20 = 0.6 \quad \checkmark \\ &[\text{since } Y = E[X]/2 + \text{Var}(X)/(2 \cdot E[X])]\end{aligned}$$

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The formula $Y = E[X]/2 + \text{Var}(X)/(2E[X])$ shows that variability always increases the expected wait above the deterministic baseline. This is the inspection paradox quantified.

c) Conditional residual life $P(Y > 3 \mid Z = 2)$:

For $X \sim \text{Uniform}[4, 16]$ and $F_X(x) = (x-4)/12$ for $4 \leq x \leq 16$:

$$P(X > x) = (16-x)/12 \quad \text{for } 4 \leq x \leq 16$$

$$P(X > x) = 1 \quad \text{for } x < 4$$

$$P(Y > 3 \mid Z = 2) = P(X > 3+2) / P(X > 2)$$

$$= P(X > 5) / P(X > 2)$$

$$= [(16-5)/12] / 1$$

$$= 11/12 \approx 0.917$$

$$[P(X > 2) = 1 \text{ since the minimum inter-arrival time is } 4 > 2]$$

There is a 91.7% chance of waiting more than 3 more minutes given the bus left only 2 minutes ago. This makes sense: since $X \geq 4$, any interval containing time t must last at least 4 minutes total, so 3 more minutes of wait is very likely when only 2 minutes have passed.